

Orthopolity: resource equalization and constrained distributions

Class-exchange dynamics, physical realizations, and empirical scope

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Abstract

Orthopolity proposes a tendency toward equal resource allocation among concentrations of resources. Its neutral allocation form can produce unequal abundances: classes whose objects require more resource contain fewer objects. Actual distributions also reflect system and medium constraints, so a departure from that form does not alone negate the proposed tendency. We formulate the neutral hypothesis and distinguish it from a claim that every distribution must be a power law. We derive a conservative class-exchange mechanism that converges to neutral allocation. An added preference produces a curved stationary profile while equalizing and constraint flows remain active and cancel. Removing the preference predicts recovery with a spectral bound. A finite-packet simulation checks the dynamics and fluctuation predictions. A separate derivation from electromagnetic energy balance gives equal throughput and equal energy in equal-thickness shells, with absorption-induced departures and causal recovery. Classical thermal radiation gives equal mean energy per field mode; quantization predicts the frequency-dependent departure $\bar{\epsilon}/(k_B T) = x/(e^x - 1)$, with $x = h\nu/(k_B T)$. An explicitly retrospective transformation of the published FIRAS monopole product connects this result to a measured spectrum, preserving its blackbody reconstruction, fitted residuals, and correlated uncertainties. The defining accounting identity depends on a reference measure. With resource cost proportional to size raised to power d , equal resource per logarithmic size interval implies a count-density exponent $d + 1$; equal resource per linear interval implies exponent d . Conservation alone selects neither allocation. Exploratory assessments of earthquake and solar-flare catalogues, an ocean biomass reconstruction and 1,300 aquatic size-spectrum estimates contradict the specified physical proxies and find aquatic slopes near the logarithmic prediction without establishing equivalence. Centred slopes, fitted dispersion and conservation cannot establish ensemble equipartition. We then test frozen forecasts, disclosing prior outcome exposure. In an engineered CPU-allocation system, independently measured non-unit costs predict allocation profiles and their response to a restriction. On an unused solar year, logarithmic neutrality predicts counts better than linear neutrality, but neutrality itself remains unresolved. In 30 size-selected algal lineages regrown in one medium, carrying capacity in biovolume is constant across a 10-fold range of cell volume, so abundance scales inversely with volume, as budget closure with a geometric cost predicts. In grazed food-web chemostats, a two-budget cost-ratio rule fails to predict pulse-driven redistribution. In directly weighed plant censuses, a flat logarithmic biomass template ranks second of five transferred models, while all three trained models predict counts better. Finite-census calibration distinguishes normalized expected stocks from expected normalized census shares; ecological neutrality remains unresolved. Resource symmetry supplies a possible physical postulate for equal expected allocations. The evidence supports conditional budget-closure forecasts under declared resource conditions and a framework for testing allocation; it does not establish neutrality across coexisting natural size classes or a new natural or cosmological law.

1. Introduction

An allocation can be equal when counted in resources and unequal when counted in objects. For example, equal biomass in successive body-mass classes requires fewer large organisms. This observation motivates the term *orthopolity* in Fabbri’s 2024 essay, following an earlier manuscript by Fabbri and Oliveira Jr. (2017). The earlier manuscript assumes uniform resource allocation across component-wealth values; the later essay proposes a much broader natural and cosmological principle. The useful question is which allocation statement can be defined and tested independently of the abundance distribution it seeks to explain.

The proposed tendency must be distinguished from the resulting distribution. A standing tree experiences gravity while its structure and anchorage support it. An apple thrown upward initially rises while gravity acts; catching it or lodging it against a ceiling introduces an interaction that can prevent its return to the ground. These outcomes reflect gravity together with other interactions and initial conditions. The same logical distinction guides orthopolity: its visible signature can be altered or obscured by the construction of the system and its medium. Human age and height distributions motivate analyses of such constraints. Their lack of an apparent neutral signature would not, by itself, establish a general absence of evidence for orthopolity in Nature. Section 2.5 states how this distinction enters the research programme.

The ecological precedent is substantial. Sheldon, Prakash and Sutcliffe (1972) studied ocean particle size distributions; Gaedke (1993) explicitly related equal biomass in logarithmic classes to normalized biomass spectra; Hatton et al. (2021) reconstructed the ocean spectrum from bacteria to whales. Arranz et al. (2022) studied variation among hundreds of lake fish communities and its ecological correlates. Variation among individual systems is therefore already a research topic. Energetic equivalence (Damuth, 1981) is related, but its species-population sampling unit differs from community abundance per size interval. Species richness across sizes prevents automatic interchange of those statements.

Power laws arise under different generative mechanisms, and apparent straight lines in logarithmic plots do not establish a power-law distribution (Newman, 2005; Clauset, Shalizi and Newman, 2009). Rewriting a distribution as inverse resource cost does not identify a new mechanism. We specify the resource and comparison measure, derive an explicit equalizing mechanism and its constrained response, and work out physical realizations from electromagnetic transport and thermal radiation. In the latter, independently counted modes define the comparison measure, and quantization determines a departure with a controlled classical limit. We then assess retained empirical cases and frozen forecasts with their exposure history. These contributions distinguish a proposed principle, its governing assumptions, and its observable expressions. The accounting identity, reversible transport mathematics, and ecological biomass equivalence have established precedents.

2. Mathematical formulation

2.1 Objects, resources, and a reference measure

Let k be a positive size coordinate on a finite domain $D = [a, b]$, with $0 < a < b < \infty$. Choose a reference measure μ , with $0 < \mu(D) < \infty$, for the classes being compared. Let $n_\mu(k) = dN/d\mu$ denote object abundance and $\bar{q}(k) = E[q \mid k] > 0$ the conditional arithmetic mean of an additive resource per object. Then

$$\mathcal{O}_\mu(k) = \frac{dR}{d\mu} = \bar{q}(k)n_\mu(k). \quad (1)$$

For an observed class, the corresponding identity is simply $\sum_i q_i = N_{\text{class}} \bar{q}_{\text{class}}$. Resources may be a standing stock, such as biomass, or an accumulated event quantity over a declared observation period. The latter is not a stock resident in a size class, and neither is a flux through size classes. Additivity is needed for (1); a dynamical conservation law is not.

Define the **neutral allocation form of orthopolity relative to μ** , denoted O_μ , by

$$\mathcal{O}_\mu(k) = C > 0 \quad \text{for } \mu\text{-almost every } k \in D. \quad (2)$$

This is an assumption about the neutral allocation, rather than a claim that every constrained system has this profile. It gives $n_\mu = C/\bar{q}$. Equivalently, if an object is sampled uniformly from the population, its size density with respect to μ is

$$p_N(k) = \frac{1}{Z\bar{q}(k)}, \quad Z = \int_D \frac{d\mu(k)}{\bar{q}(k)} < \infty. \quad (3)$$

Sampling instead in proportion to the resource held gives $p_R(k) = \bar{q}(k)p_N(k)/E_N[q] = 1/\mu(D)$. Thus uniform resource-weighted size and inverse-cost object abundance are equivalent statements, under the same reference measure. This is an accounting equivalence, not a derivation of the allocation assumption. The count, resource total, and constants also satisfy $N = CZ$ and $R = C\mu(D)$. For a finite observed population, the identity is exact for class totals; a smooth density is a population model or approximation.

2.2 The measure changes the exponent

Write $u = \ln(k/k_0)$ for a fixed reference size k_0 . For $\bar{q}(k) = q_0(k/k_0)^d$, with $d > 0$, two different hypotheses give:

Equal resource with respect to	Resource spectrum	Count density with respect to dk
Linear size, $d\mu = dk$	$\bar{q}(k)dN/dk = C$	$dN/dk \propto k^{-d}$
Logarithmic size, $d\mu = du = dk/k$	$\bar{q}(k)dN/du = C$	$dN/dk \propto k^{-(d+1)}$

The second row follows because $dN/du = k dN/dk$. For biomass with $q = k = m$, the predictions are consequently density exponents 1 and 2, respectively. The 2017 manuscript's explicit uniform density over a linear wealth interval is not interchangeable with the logarithmic convention used in the empirical analyses below. Adopting logarithmic classes is a substantive specification, motivated by size-spectrum practice.

Logarithmic allocation is unchanged by changing size units or logarithm base, apart from a constant density multiplier. It also survives $z = (k/k_0)^c$ for constant $c > 0$, since $d \ln z = c d \ln k$. It is not invariant under arbitrary nonlinear coordinates. In general, $dR/dv = (dR/du)|du/dv|$. Choosing a transformation after observing R can therefore manufacture flatness; the coordinate and measure must be fixed independently.

Under $O_{\log}$ and the power-cost assumption, the slope of abundance per log interval is $-d$, the ordinary density exponent is $d + 1$, and the survival function on $[a, b]$ is

$$P(K \geq k) = \frac{k^{-d} - b^{-d}}{a^{-d} - b^{-d}}. \quad (4)$$

The familiar survival exponent d and rank-size exponent $1/d$ are approximations away from the upper boundary, not exact finite-domain identities. For an unbounded domain with $d > 0$, count

can be finite while logarithmically uniform resource has infinite total. Finite boundaries are essential to a finite-budget interpretation.

2.3 Allocation postulates and limits of the accounting

Resource symmetry as a candidate law. The proposed rationale is that, absent restrictions favouring particular resource costs, allocation should not distinguish those costs. A natural law can adopt this as a physical postulate without a deeper mechanistic derivation. One precise sufficient formulation uses K declared classes with fixed positive per-object costs q_i , allocations $R_i = q_i N_i$, and fixed total $\sum_i R_i = R > 0$. If the probability law of $(R_1, \dots, R_K)$ is invariant under exchanging allocations across the fixed classes, then symmetry and the sum give

$$E[R_i] = R/K, \quad E[N_i] = R/(Kq_i).$$

This postulate selects resource allocations as the symmetric quantities; symmetry of counts would give a different prediction. It is stronger than conservation and is not implied by an unspecified absence of constraints. Indivisible objects may require an approximation where exact exchangeability is infeasible. The class measure also matters: with cost itself as coordinate, equal resource per dq predicts $dE[N]/dq \propto q^{-1}$, whereas equal resource per $d \ln q$ predicts $dE[N]/dq \propto q^{-2}$. Transformation-based treatments of indifference likewise require a specified symmetry (Jaynes, 1968); an inferential rule does not itself establish a physical law.

Equal expectation does not establish equal snapshots or relaxation toward equality. In a divisible-resource idealization, assigning all resource to one uniformly chosen class has the required symmetry and unequal allocation in every realization. Time-average equality needs an additional condition connecting time averages to expectations or its own empirical postulate. A test must therefore specify what "tends to allocate" means and identify eligible systems through conditions independent of their observed allocation. A deeper mechanism is optional for an empirical law; the scope and predictive evidence are not.

Conservation does not imply equal allocation. On any bounded logarithmic domain, $\mathcal{O}(u) = A \exp(\beta u)$ can be normalized to the same resource total for every β . Conservation fixes the integral and leaves its shape undetermined. Likewise, maximum entropy over finitely many equally weighted object classes, constrained by a mean cost, yields $p_j \propto \exp(-\lambda q_j)$. Different reference measures or constraints give different distributions (Jaynes, 1957; Visser, 2013).

Equipartition does not imply a power law without a cost law. If $\bar{q}(k) = q_0 \exp(k/k_0)$, then $O_{\log}$ gives $dN/du \propto \exp(-k/k_0)$ and $dN/dk \propto \exp(-k/k_0)/k$. Neither is a power law. Geometric dimension supplies d only in a specified geometric model such as similar objects with volume proportional to length cubed. It does not establish the number of independent resource inputs in an arbitrary system.

Independent inputs do not generally add tail exponents. For example, two independent unit-scale Pareto variables with densities hx^{-h-1} , $x \geq 1$, $h > 0$, have product density $h^2 z^{-h-1} \ln z$, $z \geq 1$. Integration over the possible factorizations introduces a logarithmic factor. This limits the independent-input argument in the 2017 essay.

The resource must not be fitted to force the conclusion. For any positive abundance, defining $\bar{q} = C/n_\mu$ makes (2) hold identically. A test must instead specify the resource through measurements, definitions, or an independently justified cost model. An unrestricted residual multiplier called "friction" is equally uninformative unless its form or parameters are predicted independently.

2.4 What an ensemble statement would mean

Suppose a system has a power-shaped resource spectrum $\mathcal{O}_i(u) = A_i \exp(\beta_i u)$. A proposed ensemble condition $E[\beta_i] = 0$ concerns mean slopes. It does not imply $E[\mathcal{O}_i(u)]$ is constant. With equal $A_i = A$ and $\beta_i \sim N(0, \tau^2)$,

$$E[\mathcal{O}_i(u)] = A \exp(\tau^2 u^2 / 2), \quad (5)$$

which is curved whenever $\tau > 0$. Equal-total normalization changes this formula but does not make mean-zero slopes sufficient: averaging normalized spectra proportional to $\exp(cu)$ and $\exp(-cu)$ on a symmetric domain produces a nonconstant hyperbolic cosine. A median slope is a different estimand again. A claim about average resource occupancy requires spectra, a common domain, and an explicit weighting of systems; slopes alone do not identify it.

More generally, let $H(u) = \sum_{i=1}^n w_i A_i e^{\beta_i u}$ on a common log-domain, with fixed positive weights and amplitudes. Then

$$H''(u) = \sum_i w_i A_i \beta_i^2 e^{\beta_i u} \geq 0.$$

If any slope is nonzero, the derivative is strictly positive, so the arithmetic mixture cannot be constant on an open interval. Exact flatness requires every contributing slope to be zero. With $p_i(u) = w_i A_i e^{\beta_i u} / H(u)$, its log-curvature is $d^2 \ln H / du^2 = \text{Var}_{p(u)}(\beta)$. Per-system total normalization preserves these conclusions. They are standard convexity consequences for exact power spectra; arbitrary curved profiles can cancel to a flat mean. They therefore neither exclude a general resource-symmetry law nor convert noisy fitted slopes into a profile test. The scientific assessment gives a quantitative bound for approximate flatness and its weighting restrictions.

Mean and variance also do not require a Gaussian distribution. Gaussianity is an additional maximum-entropy modelling choice, and selecting it by AIC is not a goodness-of-fit test. If $\beta_i \sim N(\mu_\beta, \tau^2)$ and reported slope error is independent $N(0, \sigma_i^2)$, the predicted probability that the *reported slope* lies in $[-c_i, c_i]$ is

$$\Phi_N\left(\frac{c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right) - \Phi_N\left(\frac{-c_i - \mu_\beta}{\sqrt{\tau^2 + \sigma_i^2}}\right). \quad (6)$$

The expression $2\Phi_N(c_i/\tau) - 1$ applies to zero-centred *latent* slopes. Comparing it to observed estimates without their errors is a different calculation. Fitting τ on the same slopes makes either comparison an internal diagnostic, not independent validation.

There is a valid simultaneous-resource restriction for an individual system: $\mathcal{O}_1/\mathcal{O}_2 = \bar{q}_1/\bar{q}_2$. Both spectra can be exactly flat only if their mean resource ratio is size-independent. Across systems, however, $\beta_{1i} - \beta_{2i} = d_{1i} - d_{2i}$ for power cost laws. Equal variances and correlation one follow only if that difference is constant and slope variance is nonzero, whether or not equipartition holds. They are not a general test of ensemble equipartition.

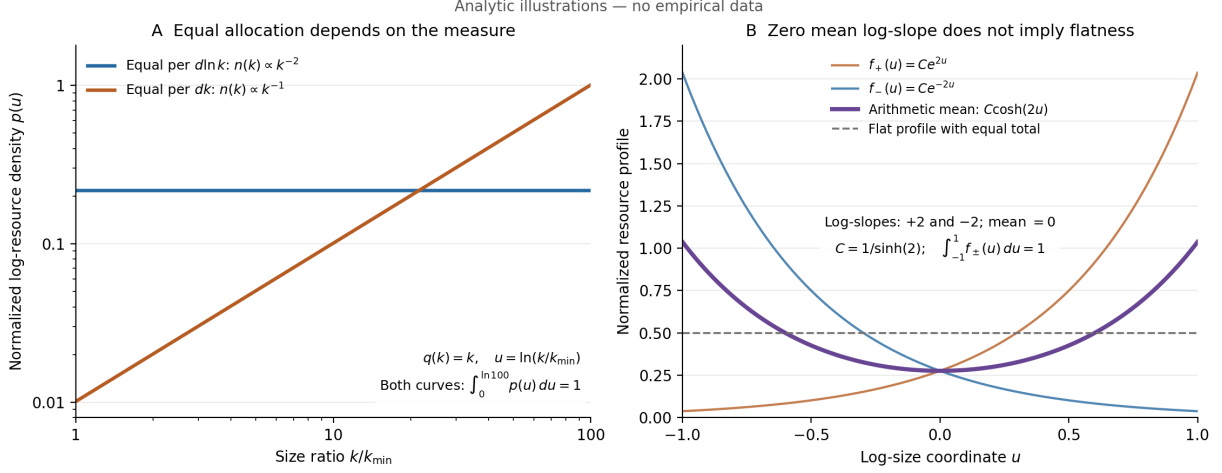


Figure 1. Analytic illustrations, not empirical data. (A) With resource equal to size, linear and logarithmic resource equipartition produce different count exponents. Both curves show unit-total resource density in the logarithmic coordinate. (B) Two unit-total resource profiles on a common domain have opposite log-slopes and therefore zero mean log-slope, but their arithmetic mean is not flat. Generated by `run_theory.py`; a vector version is available.

2.5 System constraints and observable distributions

Three levels of description must be kept distinct: an underlying resource-equalizing tendency, the neutral signature (2), and the outcome of the complete system. The latter depends on the proposed tendency together with resource costs, other processes, forcing, and initial and boundary conditions. A model containing an equalizing contribution can therefore predict a non-neutral stationary state or a transient with no clear neutral signature. Establishing such a contribution requires an explicit model or discriminating observations; equation (1) alone does not identify it.

Age and height distributions are motivating examples of constrained outcomes. A worked age-distribution account must specify birth history, survival, and migration; a height account must specify developmental processes, biological variation, and environmental conditions, as well as the population sampled. An orthopolity application must additionally identify the resource, its concentrations, and the comparison measure. No completed orthopolity model of either example is supplied here. The relevant task is to derive the combined outcome and explain how its form depends on the constraints.

A failure of the neutral prediction is a finding about that prediction in the stated setting. It does not automatically establish the absence of the broader tendency, and counting distributions without its visible signature does not establish a general absence of evidence in Nature. A constrained model has its own predictions, which can succeed or fail. Specify the constraint through physical reasoning or independent measurements, derive its effect on the profile, and test the predicted response when it changes. Where recovery of neutrality is predicted, specify its conditions and timescale. This permits both neutral and non-neutral outcomes to inform the theory, while preserving the distinction between an observed departure, a proposed cause, and a tested explanation.

2.6 A mechanism on resource-concentration classes

Consider K fixed size classes, with positive reference widths w_i , fixed per-object resource costs q_i , nonnegative class stocks R_i , and conserved total $B = \sum_i R_i$. The classes represent populations of resource-bearing objects; the state is their aggregate resource. Equivalent counts $N_i = R_i/q_i$

may be continuous or expected counts, and their total need not be conserved. This effective model does not specify individual-object assembly or demographic kinetics. The full derivation states its interpretation and assumptions.

Let $g_{ij} = g_{ji} \geq 0$ be conductances on a connected undirected graph. Neutral exchange has current $J_{ij}^0 = g_{ij}(R_i/w_i - R_j/w_j)$ and dynamics $\dot{R}_i = -\sum_j J_{ij}^0$. Positivity follows from nonnegative jump rates g_{ij}/w_i ; antisymmetry of currents conserves total resource. Connectedness makes the stationary resource density constant, giving

$$R_i^0 = \frac{Bw_i}{\sum_j w_j}, \quad N_i^0 = \frac{Bw_i}{q_i \sum_j w_j}.$$

This is a mechanism selecting equal resource per declared class measure. The local exchange symmetry and the measure are assumptions of the model; they are not supplied by conservation alone. To model a constraint, prescribe dimensionless class preferences V_i and add downhill transitions:

$$a_{ij} = \frac{g_{ij}}{w_i} \exp(\max\{V_i - V_j, 0\}).$$

The rate a_{ij} is from i to j . Its neutral part g_{ij}/w_i remains, and the additional part is nonnegative. With column generator $A_{ji} = a_{ij}$ for $j \neq i$, $A_{ii} = -\sum_{j \neq i} a_{ij}$, define

$$\pi_i = \frac{w_i e^{-V_i}}{Z}, \quad Z = \sum_j w_j e^{-V_j}.$$

Then $\pi_i a_{ij} = \pi_j a_{ji}$ on every edge. Detailed balance selects the unique stationary resource $R_i^V = B\pi_i$. This uses standard reversible-chain mathematics (Levin and Peres, 2017), specialized here to fixed concentration classes and an explicit additive constraint mechanism. At that stationary state the neutral edge current is

$$J_{ij}^0 = \frac{Bg_{ij}}{Z}(e^{-V_i} - e^{-V_j}), \quad J_{ij}^c = -J_{ij}^0.$$

Thus a nonuniform stationary distribution can have an active equalizing contribution balanced by the constraint. Setting V to a constant removes the added rates and lets the actual constrained allocation relax toward neutrality.

For $p = R/B$, set $E^2 = \sum_i (p_i - \pi_i)^2 / \pi_i$. Writing $z_i = p_i / \pi_i$ gives

$$\frac{dE^2}{dt} = -2 \sum_{i < j} \pi_i a_{ij} (z_i - z_j)^2 \leq -2\lambda E^2,$$

where $\lambda > 0$ is the smallest nonzero eigenvalue of the symmetric matrix $-D_\pi^{-1/2} A D_\pi^{1/2}$. Consequently $E(t) \leq E(0)e^{-\lambda t}$ and $\text{TV}(p, \pi) \leq E/2$. The same bound applies after release, using the neutral generator and the state actually present at the switch. If connections are removed, each component retains its own resource total and only component-wise equalization follows.

The simulation fixes 24 equal-log size classes on $[1, 64]$, $q_i = k_i^{1.5}$, and adjacent conductances $1/\Delta \ln k$. A prescribed quadratic preference in $\ln k$, centered at $k = 8$ with width 0.75, produces a curved resource profile. The corresponding continuous count form is $dN/dk \propto k^{-2.5} e^{-V(\ln k)}$, a truncated lognormal shape for this quadratic choice. Each phase runs for eight relaxation times. The neutral and constrained spectral gaps are 0.569805 and 1.991712 per model time unit. The final relative errors in the neutral, constrained, and released phases are 4.73×10^{-4} , 2.41×10^{-7} , and 5.51×10^{-11} , each below its predicted bound. The opposing stationary drift vectors cancel to Euclidean residual 3.22×10^{-15} .

There are 4,096 equal-resource packets in each of 128 realizations, initially all in the first class. Exact matrix-exponential checkpoint transitions preserve the state at both switches. Packets remain independent, giving class-count law $X(t) \sim \text{Multinomial}(4096, p(t))$ and share covariance $(\text{diag}(p) - pp^T)/4096$. Uneven finite snapshots are therefore predicted even at neutral mean allocation. Rates, costs, preferences, and predictions were fixed before drawing the stochastic trajectories. The computational results and checks are retained with the model report; the simulation supplies a constructive example, not new natural-system observations.

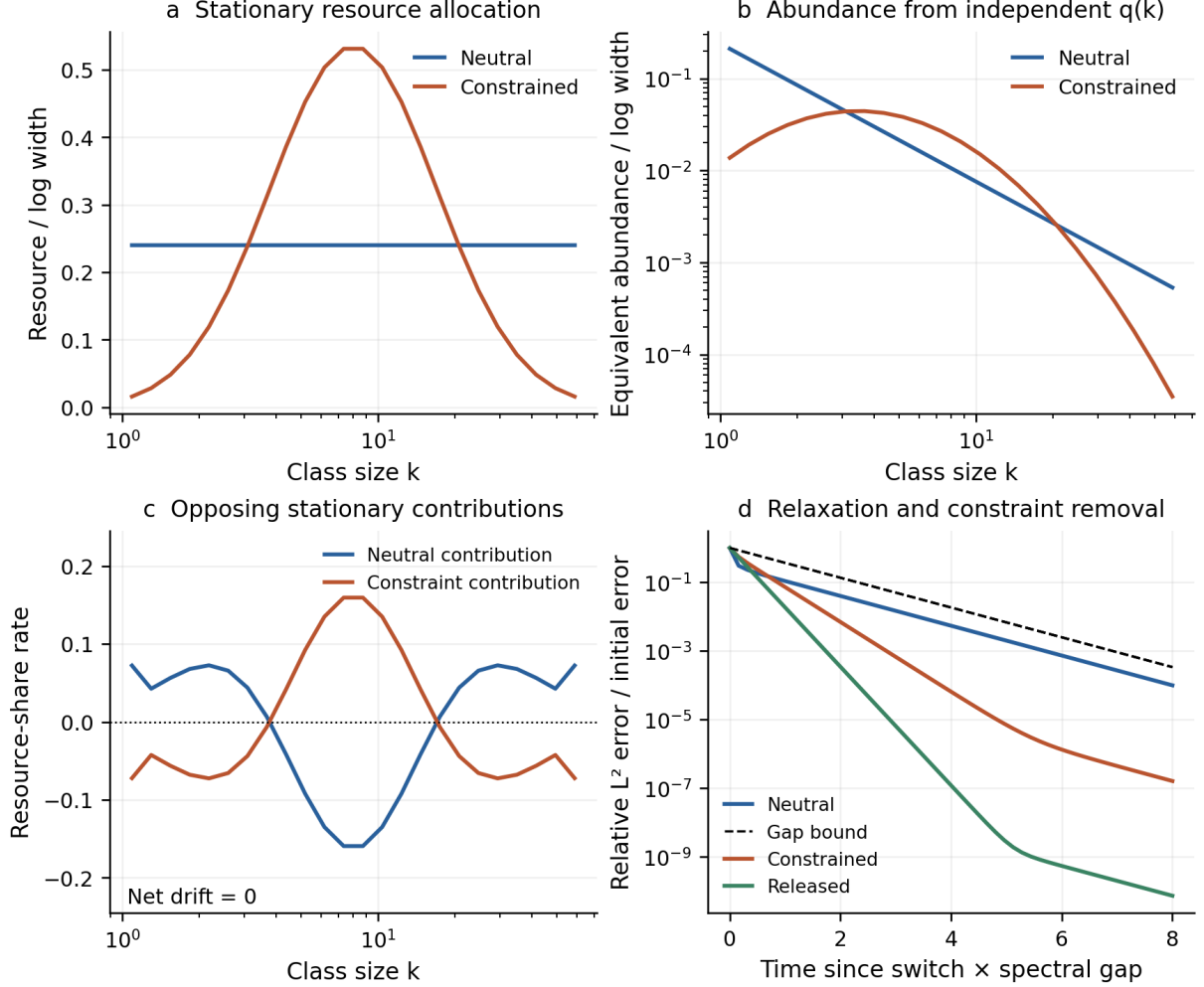


Figure 2. Fixed class-exchange model. Equalizing kinetics select neutral resource shares; an added class preference selects a curved profile and changes equivalent abundance. At constrained equilibrium the neutral and added contributions cancel. Removing the preference preserves the current state and predicts recovery under the neutral dynamics. The displayed convergence curves are deterministic expectations; stochastic packet realizations and their uncertainty are retained in the linked report.

2.7 A physical realization from electromagnetic transport

For electromagnetic energy density u and Poynting vector $\mathbf{S}$, Maxwell's equations give $\partial_t u + \nabla \cdot \mathbf{S} = -\mathbf{J} \cdot \mathbf{E}$ (Poynting, 1884; Haus and Melcher, 1989). Define $P(r) = \int_{S_r} \mathbf{S} \cdot \mathbf{n} dA$ outside a source enclosed by an inner sphere r_0 with steady power $P(r_0) = L$. Integrating the balance over a transparent, source-free shell with no net energy accumulation gives $P(r_2) = P(r_1) = L$. Isotropy is a separate condition: it makes the local radial flux density $F = L/(4\pi r^2)$; otherwise this is its sphere average.

Under outgoing radial transport at constant speed c , without scattering or secondary emission in the modeled component, a specified absorber with coefficient $\kappa(r)$ gives

$$\frac{dP}{dr} = -\kappa(r)P, \quad P(r) = L \exp\left[-\int_{r_0}^r \kappa(s) ds\right].$$

This is the absorption law specialized to spherical transport (Condon and Ransom, 2016). An absorbing layer with optical depth $\ln 2$ halves downstream power. The missing radiative power is deposited in matter, so the modeled radiation component has an open budget. Its departure follows from the prescribed medium and the same energy balance.

These nested surfaces receive the passage of the same energy; their powers cannot be summed as disjoint resource shares. Disjoint shell stocks do give an additive comparison: in transparent stationary radial transport, $dU = (L/c)dr$. Thus equal radial thicknesses hold equal energy, whereas $dU/d\ln r = Lr/c$ increases with radius. The resource, units, and comparison measure determine which equality is physically meaningful.

The response to removing an absorber is also determined. For sphere-averaged u , with $e = 4\pi r^2 u = P/c$, the transport equation is $\partial_t e + c\partial_r e = -c\kappa(r, t)e$. Suppose the source remains constant, the initial field is stationary under $\kappa_{\text{old}}(r)$, and absorption is set to zero throughout the modeled domain at $t = 0$. For $t \geq 0$, characteristics give

$$P(r, t) = L \exp\left[-\int_{r_0}^{\max\{r_0, r-ct\}} \kappa_{\text{old}}(s) ds\right].$$

Every radius recovers by $(r - r_0)/c$. Downstream of a layer $[a, b]$, the response starts at $(r - b)/c$ and completes at $(r - a)/c$. This is an idealized change of the absorption coefficient, with no radiation injected by the removal process. The physical derivation states the boundary history and all transport assumptions. It supplies an established physical realization of a declared equality and its constrained departure; it does not equate radial shells with organism size classes or derive abundance laws from an inverse-square flux.

2.8 Classical equipartition and quantum radiation

Thermal electromagnetic radiation supplies a second physical realization with a different reference measure: counting field modes, including polarization. For a finite set of classical normal modes in canonical equilibrium at $T > 0$, each has two quadratic terms, $H_j = P_j^2/(2m_j) + m_j\omega_j^2 Q_j^2/2$. Gaussian integration over both coordinates gives $Z_j^{\text{cl}} \propto (\beta\omega_j)^{-1}$, with $\beta = (k_B T)^{-1}$, and therefore

$$\langle H_j \rangle = -\partial_\beta \ln Z_j^{\text{cl}} = k_B T.$$

A class of M_i modes consequently has mean energy $M_i k_B T$: the resource density is equal per mode, with independently defined weight $w_i = M_i$. This is equality of equilibrium means, not identical instantaneous energies. Equilibrium preparation or thermal coupling must be justified; uncoupled harmonic modes do not thermalize each other merely because their canonical distribution can be written down. This argument uses standard equipartition and supplies no relaxation time (Tong, n.d.).

Quantization changes the allowed states. Define the resource as thermal excitation energy above the ground state, excluding the temperature-independent zero-point term. For photon absorption and emission in thermal equilibrium, photon chemical potential is zero and a mode has excitation levels $\epsilon_n = nh\nu$. The geometric sum gives

$$Z_\nu = \frac{1}{1 - e^{-x}}, \quad \bar{\epsilon}_\nu = \frac{h\nu}{e^x - 1} = k_B T f(x), \quad x = \frac{h\nu}{k_B T}, \quad f(x) = \frac{x}{e^x - 1}.$$

For $x > 0$, f decreases strictly from 1 toward 0. Its limits are $f(x) = 1 - x/2 + x^2/12 + O(x^4)$ near zero and $f(x) \sim xe^{-x}$ at large x . The quantum level spacing relative to thermal energy specifies the departure before any spectrum is observed. Raising temperature or lowering frequency recovers classical relative equality. This is a restriction of the available energy states; it does not establish the additive decomposition into opposed classical currents proved in Section 2.6.

In a large three-dimensional vacuum cavity, counting two polarizations gives the mode density $dM/(Vd\nu) = 8\pi\nu^2/c^3$. Multiplication by $\bar{\epsilon}_\nu$ yields the Planck energy density and, for isotropic radiation, the spectral radiance:

$$u_\nu(T) = \frac{8\pi h\nu^3}{c^3(e^{h\nu/(k_B T)} - 1)}, \quad B_\nu(T) = \frac{c}{4\pi}u_\nu(T).$$

Thus the classical equality per mode gives $u_\nu^{\text{cl}} \propto \nu^2$ and $dU/(Vd\ln\nu) \propto \nu^3$, neither a flat linear-frequency nor a flat log-frequency spectrum. If photons are the counted objects, $q(\nu) = h\nu$ and their mean number per mode approaches $k_B T/(h\nu)$ in the classical regime. Extending classical equipartition to infinitely many modes gives divergent total energy; quantization makes it finite, $U/V = 8\pi^5 k_B^4 T^4/(15h^3 c^3)$. These are established results (Planck, 1901; Tong, n.d.; Condon and Ransom, 2016), here specifying the resource, comparison measure, and constraint in the allocation framework.

2.9 Connection to the measured thermal spectrum

FIRAS measured the cosmic microwave background against calibrated blackbody references and separated monopole, dipole, and Galactic components (Fixsen et al., 1996). NASA LAMBDA’s distributed monopole table contains a 2.725 K blackbody plus the residuals from that published analysis, with marginal uncertainties and a Galactic model column. The residuals already follow temperature and foreground fitting. The table therefore supports a transparent re-expression of the published result; agreement with its embedded template cannot be counted as a new independent confirmation.

Our retrospective calculation retains all 43 rows and fixes $T_0 = 2.725$ K from the product definition. With wavenumber $\tilde{\nu}$ in inverse centimetres, $\nu = 100c\tilde{\nu}$. Radiance remains per hertz, with $1 \text{ MJy/sr} = 10^{-20} \text{ W m}^{-2} \text{ Hz}^{-1} \text{ sr}^{-1}$. Dividing the inferred energy density by the independent mode density gives

$$\frac{\hat{\epsilon}_\nu}{k_B T_0} = \frac{c^2 I_\nu}{2\nu^2 k_B T_0} = \frac{I_\nu}{B_\nu^{\text{RJ}}(T_0)}, \quad B_\nu^{\text{RJ}} = \frac{2\nu^2 k_B T_0}{c^2}.$$

The channels span 68.05–639.46 GHz and $x = 1.199$ –11.262. Their transformed mode energies range from 0.518 to 1.32×10^{-4} of $k_B T_0$ (Figure 3). They illustrate the quantum departure, not the $x \ll 1$ classical plateau. The CMB can retain a Planck spectrum through collisionless cosmological redshifting; this example does not require current thermal contact with surrounding matter (Condon and Ransom, 2016, Section 2.6.2).

For residual diagnostics we use the source’s original residual column, not the difference between rounded monopole intensities and a newly calculated curve. The source’s approximate frequency covariance is $C_{ij} = \sigma_i \sigma_j Q_{|i-j|}$, using its 43 published lag coefficients (Fixsen et al., 1996, Section 3.3). The matrix is positive definite. Applying the linear radiance-to-mode conversion to both axes propagates this covariance to the mode-energy ratios. The residual quadratic form is $r^T C^{-1} r = 49.774$; it is a descriptive audit of the published approximation, with no new fit or goodness-of-fit probability. It need not reproduce the original full-analysis statistic, 46 for 40 degrees of freedom. Diagonal inverse-variance weighting gives residual RMS 18.90 kJy/sr, or

49.28 parts per million of the tabulated peak radiance. This defined amplitude summary does not assume that the channels are independent.

Using exact SI constants at the rounded source coordinates gives a discrepancy $I_{\text{posted}} - [B_\nu(T_0) + r_{\text{published}}]$ of -0.127 to $+7.598$ kJy/sr, with radiances in common units. This discrepancy is retained without assigning a cause or replacing the original residuals. Absolute calibration, foreground modeling, and the prior fits remain inherited from the published analysis. The thermal-radiation report retains the full derivation, source bytes, covariance transcription, and reproducible calculation. This example connects a physical equality and an independently derived quantum departure to established observational evidence under its documented conditions.

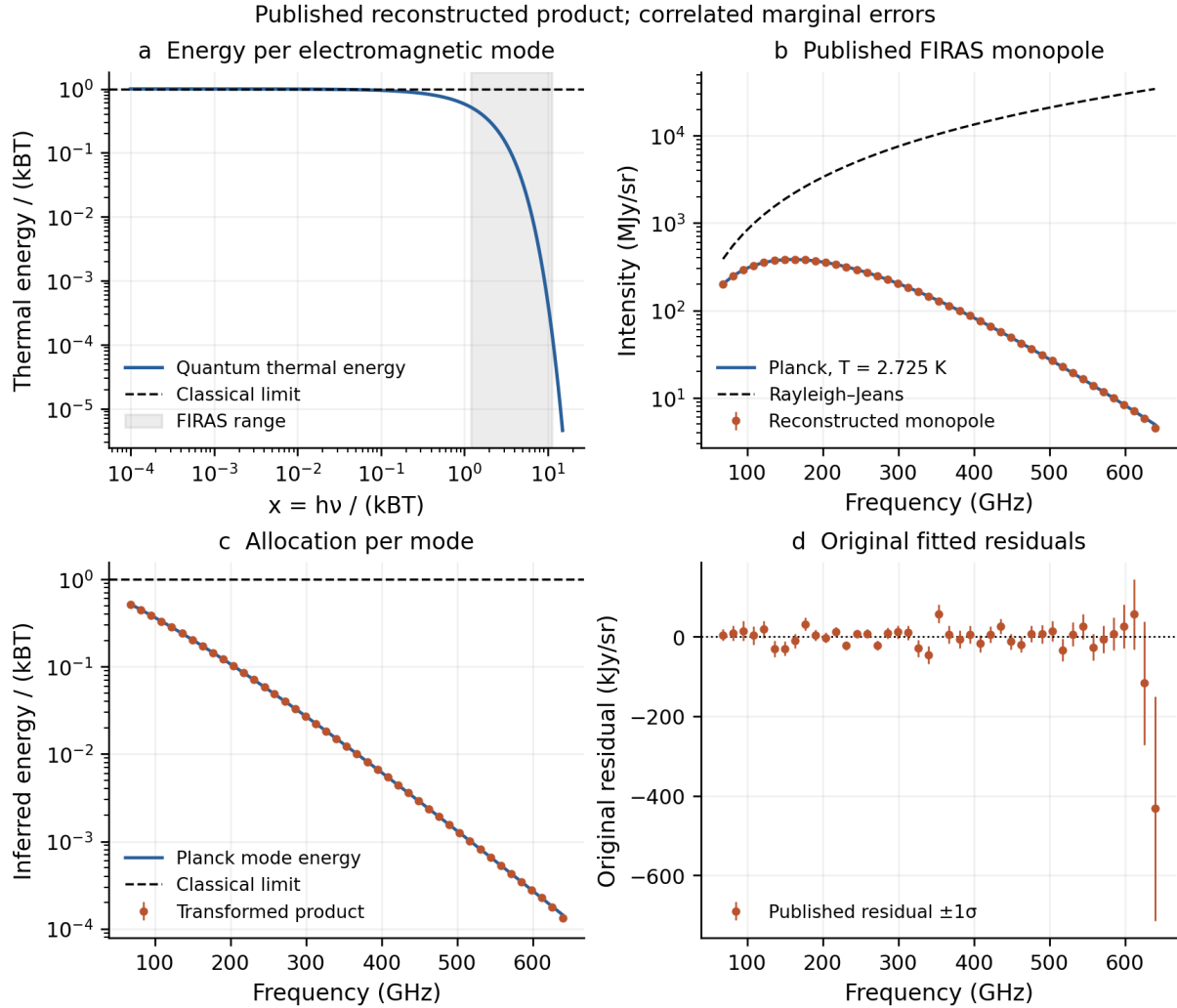


Figure 3. Thermal energy per electromagnetic mode. (A) The theoretical quantum factor $f(x)$ approaches classical equality as $x \rightarrow 0$. (B) The NASA FIRAS reconstructed monopole product and Planck and Rayleigh-Jeans radiances at the product's 2.725 K reference temperature. (C) The same product expressed as energy per mode relative to $k_B T_0$. (D) Original published residuals with marginal one-sigma uncertainties; channels are correlated. The Planck template is part of the distributed product, so panels B and C illustrate the published result rather than provide a new independent model test. All 43 channels are retained.

3. Data and methods

3.1 Sources and scope

Data source	Analysed material	Resource and size coordinate
NOAA GOES XRS flare reports, 2022–2024	10,501 catalogue events; 1,168 paired events in the primary evaluation domain	End fluence in J/m ² ; peak irradiance in W/m ²
USGS ComCat, 2010–2024	6,639 events after retaining moment-magnitude types and rounding to 0.1 at threshold 5.5	Magnitude-derived energy proxy as both resource and coordinate
Hatton et al. (2021)	23 one-decade biomass bins, upper 200 m and full water column	Body mass for both
GLOSSAQUA (Ersoy et al., 2025)	1,300 normalized biomass-spectrum estimates from 16 study identifiers	Body mass for both

These are four data sources; the two ocean domains are related reconstructions. GLOSSAQUA contains published estimates rather than organism-level observations. Study identifiers do not guarantee independent authors, methods, sites, or measurement errors. The sources are convenience cases for evaluating a hypothesis, not a representative sample of natural systems. Input bytes are fixed by SHA-256 manifests; provenance and source notices accompany the data.

3.2 Analysis-plan provenance

The pilot configuration explicitly labels itself exploratory. The subsequent aquatic/ocean plan is retained as prereg_2026-09-09.json. In local history, commit 6df364f contains the plan and raw slope data, preceding analysis commit a0c9973. This records an order of commits. It does not establish an immutable public registration, independently verified blinding, or when values were first viewed. Statements of blinding in the historical file are author reports. The current corrections are retrospective, and the later ensemble, stratification, and variance analyses are exploratory. No claim of confirmatory preregistration is needed for the descriptive and mathematical results here.

3.3 Accounting and uncertainty

For bins B_j of logarithmic width w_j , compute $\hat{O}_j = \sum_{i \in B_j} q_i / w_j$ and $\hat{C} = \sum_j \sum_{i \in B_j} q_i / \sum_j w_j$. The normalized profile $\phi_j = \hat{O}_j / \hat{C}$ has width-weighted mean one by construction. This is not evidence of flatness. Empty bins remain in the profile, the final boundary is included, and missing resources require an explicit exclusion policy.

A factor- F endpoint drift across log-domain width L corresponds, for a power-shaped spectrum, to slope tolerance $c = \ln(F)/L$. This differs from requiring every $\phi_j \in [1/F, F]$: the latter permits a maximum-to-minimum ratio of F^2 . Both slope and profile deviations matter. A slope equivalence interval combined with a point-estimate profile screen is not full simultaneous equivalence; that would require uncertainty for the whole profile. Failure to establish equivalence is not, by itself, evidence of a nonzero departure.

For flares, the arithmetic mean resource exponent is fitted by Gamma quasi-likelihood on 2022 data, with bounded count-density fitting on 2023–2024 over $[10^{-5}, 10^{-3}]$ W/m². Log-resource least squares would target a different conditional quantity unless an appropriate retransformation were justified. Pilot intervals use 600 calendar-month block resamples for flares and calendar-year

blocks for earthquakes. These do not capture all dependence or selection bias. The temporal split separates estimation samples but does not make the exploratory domain choice a prospective prediction.

Power-law diagnostics use maximum likelihood and a refitted parametric-bootstrap KS statistic on declared support, following the approach of Clauset et al. (2009). The rounded earthquake magnitudes are tested on their integer grid: for $K = (M - M_0)/\Delta$, $P(K = k) = (1 - r)r^k$ with $r = 10^{-b\Delta}$. Continuous-reference KS calibration is unsuitable for these ties. Parametric-bootstrap probabilities remain conditional on an independent-event working model; catalogue clustering is a limitation.

3.4 Aquatic estimates and reconstruction uncertainty

The primary GLOSSAQUA subset requires body mass on the size axis, finite slopes and positive ordered size limits, and the normalized biomass-spectrum convention. For biomass per unit linear mass $B(m) \propto m^t$, equal biomass per log mass requires $t = -1$. We examine $\beta = t + 1$ and each estimate's reported size span. The historical plan uses $F = 1.25$ and $F = 2$, a 2,000-replicate study-block bootstrap for the pooled median, and a joint criterion involving its interval and the fraction of slopes inside their tolerances. That fraction ignores individual slope uncertainty and must not be interpreted as formal individual equivalence. Slope summaries cannot detect within-spectrum curvature. A post-audit sensitivity explicitly excludes StudyID_07 from range-dependent assessment because its reported bounds are physically implausible; no replacement bounds are imputed (§4.4).

For the ocean reconstruction, 20,000 simulated profiles propagate reported multiplicative uncertainty factors using lognormal draws centred on the published log estimates, with log-scale standard deviation $\ln(f)/1.96$. The published bounds match estimate divided and multiplied by f . Independent errors across groups and bins are a working assumption, not a property established by those bounds. The reconstruction is not a sample of independent organisms, and its plateau was selected after inspecting the upper-ocean estimates.

Exploratory random-effects and latent-distribution fits use reported errors where available. Intervals converted to standard errors assume their advertised coverage and approximate normality. Size-spectrum estimators can have biased estimates and miscalibrated intervals (Edwards et al., 2017, 2020). Repeated spectra within studies and estimated error variances further limit inference; independence-based pooled intervals are not primary evidence of a population mean.

4. Results

4.1 Earthquakes contradict the specified energy-proxy allocation

Gutenberg–Richter scaling $N(\geq M) \propto 10^{-bM}$ and proxy $E \propto 10^{\gamma M}$ give resource per log energy proportional to $E^{1-b/\gamma}$. Flatness therefore requires $b = \gamma$. At threshold 5.5, the fitted $b = 0.998$ has year-block 95% interval $[0.973, 1.024]$, far below the chosen $\gamma = 1.5$. Thresholds 6.0 and 6.5 give 0.984 and 0.977. Using conversion exponent 1.44 also leaves a substantial discrepancy.

At threshold 5.5 the discrete KS statistic is approximately 0.0077, with a parametric-bootstrap probability about 0.26. This is non-rejection of the fitted geometric model, not proof that the model is exact. A power-like tail can thus coexist with a resource-proxy allocation that disagrees with $O_{\log}$. No discrete lognormal comparison was conducted. Magnitude-derived energy is not independently measured radiated energy; the finding is restricted to the specified proxy and catalogue, without declustering or a regional completeness model.

4.2 Flares disagree with the joint prediction on the selected domain

The fitted resource exponent is $d = 0.858$ with 95% interval $[0.697, 1.054]$. The implied density exponent is 1.858, compared with 2.239 $[2.085, 2.399]$ in the paired evaluation sample. The estimated gap is 0.382 $[0.125, 0.620]$. A rise-phase fluence sensitivity gives gap 0.411 $[0.208, 0.604]$. This alternate resource has no missing values in the domain, but does not prove the primary end-fluence analysis free of missingness bias.

For all 1,306 in-domain evaluation events, peak irradiance has fitted exponent 2.274 and KS statistic about 0.032; its bootstrap probability is about 0.02. The power law is therefore inadequate under this working test. Its likelihood cannot clearly distinguish it from the fitted lognormal. Exponent mismatch tests the conjunction of equipartition, power-shaped mean cost, transfer between years, and the count model; it cannot isolate equipartition from all those assumptions.

The gaps at four lower cutoffs are -0.480 , 0.191 , 0.382 , and 0.960 . This substantial sensitivity discourages treating one selected range as a universal scaling regime. Fluence is instrument-band irradiance integrated over an event, not total flare energy.

4.3 The ocean reconstruction is compatible with a broad, uneven plateau

The upper-ocean summary reproduces the published abundance slope at approximately -1.039 . Its maximum-to-minimum biomass ratio is 38.8 across all 23 bins, and 1.7 within the selected plateau of 16 bins. The corresponding full-water-column plateau ratio is 4.3. The 23 one-decade bins span 23 decades between outer edges; their centres are 22 decades apart. Similarly, the plateau covers 16 decades between edges and 15 between centres.

Under the stated uncertainty propagation, only 2 of 23 upper-ocean bin intervals lie wholly outside the factor-1.25 band; the corresponding full-column count is 3. These are pointwise intervals without multiplicity adjustment. The remaining intervals can include both near-flat and materially non-flat values, so their overlap with the band establishes neither equality nor absence of departures. Correlated reconstruction errors and post hoc range selection further limit the assessment. These results re-express Hatton et al.’s reconstruction; they do not independently replicate it.

The full upper-ocean range has a fitted biomass log-slope of -0.0392 and a propagated 90% interval approximately $[-0.058, -0.026]$, outside the factor-1.25 slope tolerance of 0.00421. Thus the assumed uncertainty model indicates a systematic gradient even though most individual bin intervals overlap the pointwise band. Pointwise overlap does not make the whole-profile question uninformative; its interpretation remains conditional on the reconstruction error model.

4.4 Aquatic slopes are close in location, without established equivalence

Among 1,300 normalized biomass spectra from 16 study identifiers, the pooled median slope is -1.015 . The study-block 95% interval for its departure from -1 is $[-0.100, 0.010]$. The median slope tolerance is 0.0317 at $F = 1.25$ and 0.0984 at $F = 2$. This interval is not wholly within either band. Under the historical joint criteria, the result is **not supported**.

The reported-slope fractions inside their individual drift tolerances are 0.07846 ($F = 1.25$) and 0.24231 ($F = 2$), counting boundary cases inclusively. These are descriptive compatibility fractions. Their complements are not estimates of the fraction of true spectra violating equipartition, because measurement error and within-spectrum shape are not accounted for.

These historical fractions also depend on a metadata defect: all 377 StudyID_07 records report bounds of 2×10^{-8} to 2×10^{27} pg C, a 35-decade range with a physically implausible upper body

mass of 2×10^{12} kg C. Excluding those records from this range-dependent calculation leaves 923 slopes from 15 studies. Their point-slope compatibility fractions are 9.32% at $F = 1.25$ and 28.93% at $F = 2$. The source-derived replacement bounds are unknown; the original slopes remain in the location summary. The remaining records also require a methods and units audit. Consequently neither version of the pass fraction should be treated as a validated estimate of ecological prevalence.

Two studies supply 1,016 of the 1,300 estimates, approximately 78%. A central estimate near -1 is worth investigating, but it neither establishes a mean of -1 in a defined population nor establishes equal average biomass occupancy. The stronger ensemble interpretation fails on both statistical and mathematical grounds (§2.4).

4.5 Exploratory diagnostics identify further limits

Latent-shape fits and random-effects summaries describe appreciable variation in the error-reporting subset. Their fitted dispersion includes ecological differences, study methods, and potentially unmodelled dependence. An I^2 estimate is not a measured fraction of variance caused by ecosystem differences. AIC rankings among Gaussian, Laplace, and Student families do not establish adequacy or identify ecological classes. Error-aware, matched-sample pass fractions are internal model checks, not independent predictions of individual failure.

The cross-study comparison of 639 lake-fish estimates with 377 records associated with Gaedke’s Lake Constance study does not partition spatial and temporal variance. Different taxa, sampling designs, fitting methods, and measurement uncertainties prevent treating their variance ratio as a bound on one population. Reproducing Arranz et al.’s reported summary statistics from the same underlying observations checks extraction, not independent replication of a dispersion parameter.

Exponent metadata also need source verification. Counts in equal-log bins and abundance divided by linear bin width have slopes differing by one under power scaling. The Perkins et al. (2018) count-bin method illustrates why a database category alone may not identify the estimand. Choosing a mapping because it makes a slope closer to -1 or -2 would bias the test. Secondary conventions remain sensitivity analyses; a complete study-level methods audit is needed before pooling them. Checks of two dominant primary studies do not validate every record in the primary subset.

5. Frozen-forecast tests

The analyses of Section 4 assess data that had already been inspected. We therefore added tests in which the resource, its cost law, the forecasts and the decision rules are fixed before the evaluated outcomes are decoded, and in which whole units are held out: launches, a calendar year, vessels and selection treatments. Each executed study retains its inputs, frozen protocol, archived algorithms, forecasts and results in an append-only run registry. Three kinds of evidence are kept separate: measurements on engineered systems, an unused observation period of a natural catalogue, and published biological experiments reanalysed retrospectively. Published summaries of those experiments had been read, so their reanalyses are retrospective validations, not blinded tests. A further plant study uses fixed scoring and a development-only geographic split after accidental raw outcome exposure. Its freeze cannot establish that those outcomes were unseen.

5.1 Controlled allocation measurements

In a controlled workload experiment, separately measured memory and CPU costs of matrix tasks predicted completion profiles under assigned quota pairs within four tolerances frozen before 672 validation tasks (workload pilot). A fresh launch exposed a transfer boundary: the original forecast failed its unchanged tolerance in two of four conditions, and local recalibration recovered one (workload transfer). The quotas were acceptance criteria, so these tests concern the transfer of cost models, not allocation.

An allocation test requires the system, not the experimenter, to divide a resource. In a Python runtime whose global interpreter lock serializes execution, worker threads running matrix kernels of five sizes competed for one execution resource (dimensionality intervention). Separate calibration gave CPU cost degrees 1.870 and 2.686 for the two kernels. The calibrated mean-cost curves predicted complete CPU and job profiles before 16 allocation trials, with maximum CPU-share errors of 0.0035–0.0063 and job-share total-variation errors of 0.0021–0.0124. Replacing the largest-class thread with a second smallest-class thread produced the predicted CPU-share change within 0.0078. A unit-cost alternative had count errors of 0.126–0.273. This success concerns one engineered allocation mechanism; it identifies no geometric dimension and no tendency of natural systems.

5.2 Calibrated decisions and an unused observation period

Complete-profile equivalence decisions were calibrated on 180,000 independently generated datasets in 90 conditions (profile calibration). Simultaneous bands met the declared gate for dense, bounded, independent blocks at 48 blocks, with minimum coverage 93.95%, but failed at 12 and 24 blocks. With twelve serially dependent blocks, wrong-departure rates reached 43.75%. These failures define observation designs for which no neutrality verdict is available, whatever a nominal interval suggests.

For solar flares, measured rise-phase fluence costs, six logarithmic classes, three forecasts, a tolerance and an application gate were committed before the 2025 NOAA records were acquired (solar validation). Of 3,277 events, 356 met the domain rules. Logarithmic neutrality predicted the class counts much better than linear neutrality, with count total variation 0.125 against 0.573; historical counts did better still, at 0.022. The point profile departed from the $\ln 1.5$ margin, but the calibrated decision remained unresolved, as declared before acquisition, and the instrument mix changed between periods.

5.3 Published biological measurements

Study	Held out	Resource cost	and	Frozen comparison	Outcome
<i>Synechococcus</i> quotas (Harcourt et al., 2024)	Ten cultures at 25 °C	C, N and P per cell against measured diameter		Fixed cube, free power, strain means, temperature trends	Fixed cube best for C; strain means best for N and P
Size-selected <i>Dunaliella</i> (Malerba et al., 2018)	Each selection treatment	Biovolume at carrying capacity in one shared medium		Equal biovolume ($d = 1$), fitted size law, assigned exponents, equal cell number	Equal biovolume best; cells at capacity scale as $V^{-1.02}$
Food-web chemostats (Wojcik et al., 2025)	Twelve polyculture vessels	Algal N stock: biovolume times separately assayed N per volume		Persistence, development response, no-herbivore response, equal stock, two-budget cost ratio	No model beat persistence; the cost-ratio rule failed
Harvested plants (Dillon et al., 2019)	Five Colorado plots; prior raw exposure	Direct above-ground dry mass; $q(m) = m$ by definition		Log and linear neutrality, Ohio histogram, bounded Pareto and Weibull	Pareto best for biomass; trained models better for counts; ecological neutrality unresolved

Published elemental quotas of four *Synechococcus* strains tested whether a size-based cost transfers to an unused temperature (cost transfer). Trained on 16–22 °C cultures, a cubic diameter law with one fitted intercept predicted held-out 25 °C carbon quotas within a typical factor of 1.14. For nitrogen and phosphorus, strain means beat both size laws, and a fitted exponent never beat the fixed cube. Extrapolated temperature trends were worst for every element. A cost calibration can therefore transfer for one resource while failing as a size law for another resource in the same cells.

The accounting identity behind orthopolity, $Nq(V) = R$, predicts how abundance scales with per-object cost when a budget is shared. Thirty *Dunaliella tertiolecta* lineages, artificially selected for about 280 generations into small, control and large classes, were regrown separately in one F/2 medium after replete, N-deprived or P-deprived histories (size budget). The protocol was committed before any small- or large-selected outcome was decoded; each held-out selection treatment was then predicted from the other two. Across a 10.4-fold range of mean cell volume, replete carrying capacity in total biovolume was constant to within 3%, so cell number at capacity scaled as $V^{-1.02}$ (Figure 4). The frozen equal-biovolume law ($d = 1$) had held-out errors of 0.124 and 0.179 log units. It outperformed a size law fitted within part of the range, whose fitted cost dimension was 0.74 in one fold and 1.12 in the other, and an equal-cell-number law, with errors of 1.48–2.08. A carbon cost degree assigned from the *Synechococcus* study ($d = 0.91$) performed comparably; its nitrogen degree ($d = 0.80$) did not. Regrowth after N deprivation nearly restored capacity, whereas P deprivation left an overshoot of 25–31% in control and large lineages but not in small ones. Restoring the medium therefore did not guarantee a return to the replete profile.

In 24 food-web chemostats, a nitrogen pulse redistributed algal resource composition (chemostat response). Pre-pulse N and C per cell volume from separate monocultures converted three algal groups’ biovolumes into resource stocks. Forecasts for twelve held-out polyculture vessels were frozen before their post-pulse values were decoded. The pulse moved resource composition by 0.25 total variation on average, about 1.5 times the pre-pulse variation around baseline, yet no

forecast improved appreciably on persistence: 0.243 for the development response against 0.247. A two-budget allocation rule, in which a binding secondary carbon budget reweights each group's share by its measured C:N ratio, failed on three counts. Its attainable redistribution, 0.05–0.12, was below every observed departure, 0.29–0.49. Its predicted direction held in 4 of 12 vessels, the chance count. It did not beat persistence. No held-out vessel recovered its pre-pulse composition within 12 days.

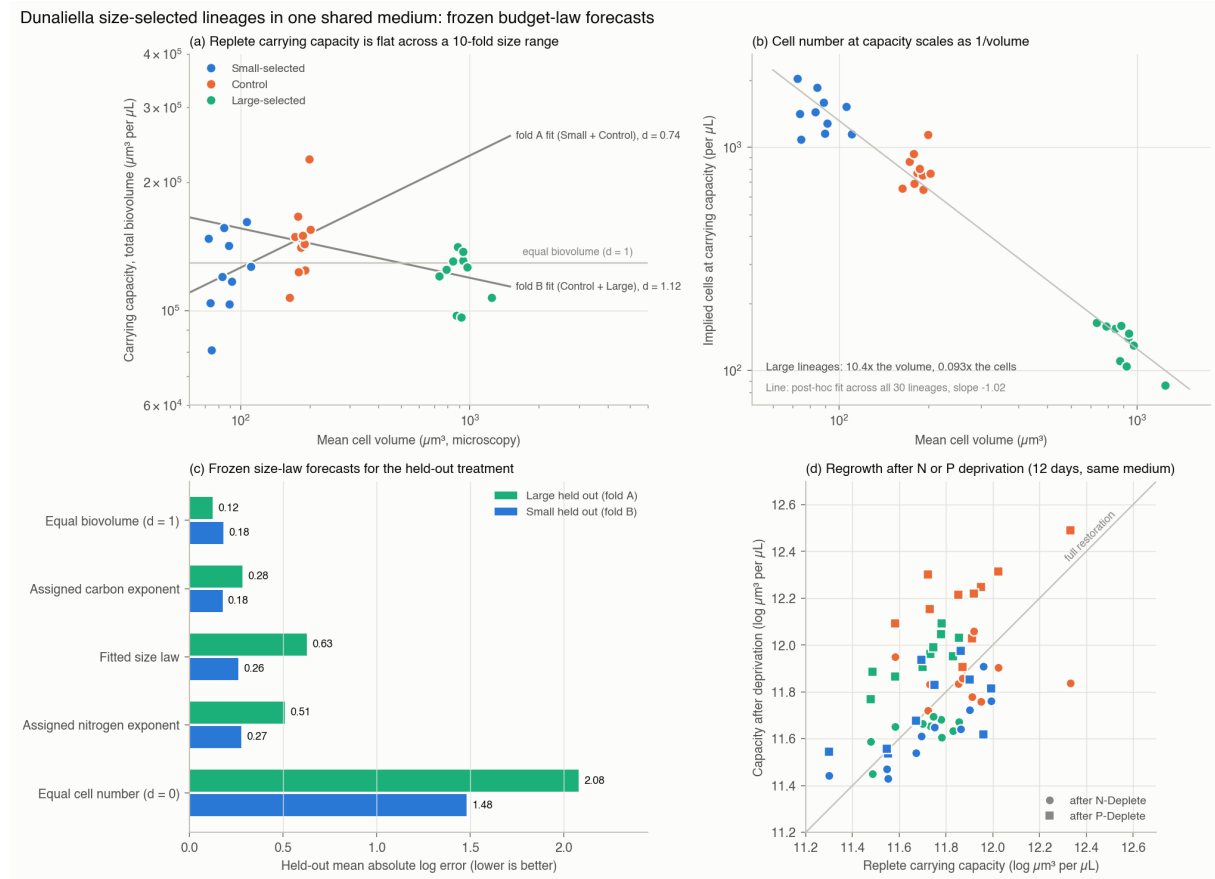


Figure 4. Size-selected *Dunaliella* lineages regrown in one shared medium, from the data of Malerba et al. (2018). (a) Replete carrying capacity in total biovolume against mean cell volume, with the two cross-fitted size laws and the equal-biovolume law. (b) Implied cell number at capacity. (c) Held-out errors of the frozen forecasts. (d) Capacity after N or P deprivation against replete capacity. Generated from retained outputs by `report_dunaliella_size_budget.py`.

5.4 Directly weighed coexisting plant stocks

Ten harvested herbaceous plots provide direct aboveground dry masses for coexisting ramets or stem clusters (Dillon et al., 2019; plant study). Forest and desert allometric masses are excluded. Five Ohio plots supply development fits; five Colorado plots supply evaluation. The protocol, algorithms and forecasts were committed before formal scoring, but raw rows had already been exposed during source inspection. This is a retrospective transfer comparison. It identifies neither a limiting nutrient nor an opportunity budget; $q(m) = m$ defines the measured stock rather than testing an independent cost law.

The frozen domain is 0.01–100 g, split into eight half-decade bins with empty classes retained. The upper bound depends only on Ohio masses. Thirty missing mass records, 144 subthreshold ramets and one above-domain ramet remain in the evaluation membership ledger. The latter weighs 112.55 g and accounts for 24.50% of one plot's known biomass. In-domain mass coverage

is 75.48% there and 99.89–99.97% in the other four plots. Scores therefore describe the declared domain; the excluded mass and unknown missing mass cannot be silently absorbed.

The primary score is equal-plot mean total variation between realized normalized biomass and each expected-stock or empirical template. Bounded Pareto scores 0.470, logarithmic neutrality 0.493, linear neutrality 0.507, the development histogram 0.510 and bounded Weibull 0.531 (Figure 5). Pareto beats logarithmic neutrality in three of five plots; large opposing plot differences leave a mean improvement of only 0.023. Count profiles provide a different comparison: the development histogram, Pareto and Weibull score 0.260–0.271, versus 0.412 for linear and 0.495 for logarithmic neutrality. Their binned count log losses are about 1.83, versus 2.08 and 2.42. These are point discrepancies, without a calibrated ecological superiority or equivalence verdict. They also do not contradict the original study’s superior within-site Weibull fits: the models here transfer across locales and share a development-fixed domain.

Synthetic calibration fixes census size and draws masses from m^{-2} on the same domain, so each bin has equal expected biomass before measurement rounding. At 160 independent ramets, mean realized stock TV is 0.441 and every simulated census has an empty bin. The largest bin’s average normalized share is 3.13%, although its normalized expected stock is 12.50%. At 1,280 ramets these values are 0.244, 87.2% and 9.12%. Thus $E[R_i / \sum_j R_j]$ differs from $E[R_i] / \sum_j E[R_j]$. A separately generated iid 95% TV envelope has exceedance rates 6.2% and 5.2%, but repeating masses in blocks of twenty raises these to 97.8% and 99.9%. This stress test is not an ecological dependence model. Field dependence, inclusion and independent regional replication remain unidentified, so the ecological neutrality decision is unresolved. An uneven small census alone cannot settle a claim about expected allocation.

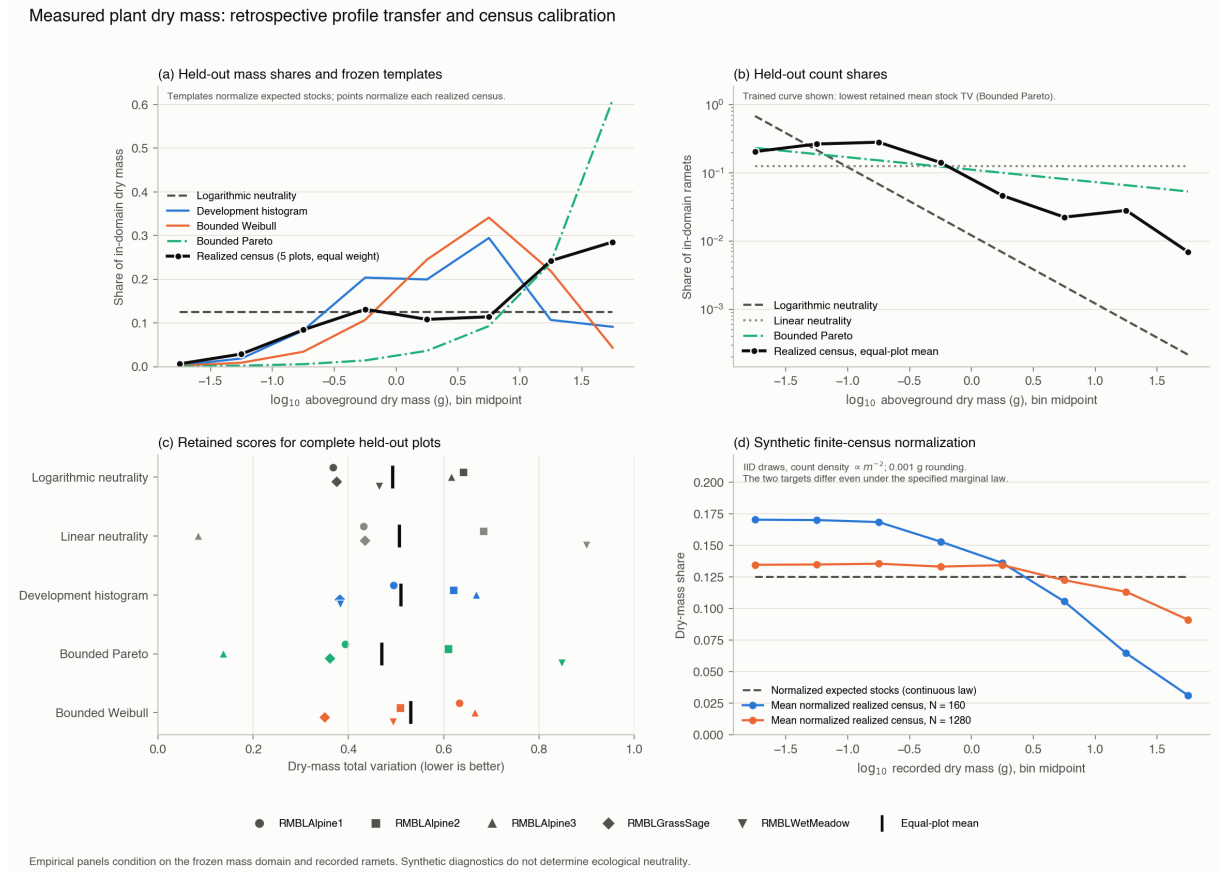


Figure 5. Directly weighed plant profiles and synthetic observation limits. (a) Equal-plot mean Colorado biomass shares and transferred stock templates. (b) Count shares and retained count forecasts. (c) Each plot’s biomass discrepancy and the five means. (d) Mean normalized census shares under iid logarithmic neutrality, compared with normalized expected stocks. All panels read retained outputs; `report_plant_biomass_profile.py` performs presentation only. Prior raw exposure and excluded biomass are retained.

5.5 What the tests establish

The positive results concern conditional budget closure: a separately measured or geometric cost, declared shared resource conditions, and abundance or resource shares predicted from them. The runtime experiment and the algal lineages are the clearest cases; in the lineages the closure holds with cost proportional to volume, so the abundance gradient carries no information beyond geometry. A rule that goes beyond closure, by predicting how a restriction redistributes resource among coexisting classes, failed in the grazed, dynamically changing food webs tested. The plant study now observes coexisting class stocks directly, but it does not establish equal expected allocation: the flat template is not best for biomass or counts, and the ecological observation law is uncalibrated. No natural size-class study in Section 5 has independently identified neutral eligibility and a design sufficient for the expected-allocation verdict. That remains the decisive missing test.

6. Discussion

The conditional identity is useful because it forces a researcher to name the resource, sampling unit, coordinate, and reference measure before interpreting a slope. It also separates a distributional question from an allocation question: a power-law model can fit without the specified resource being equally allocated. Conversely, equal allocation can hold with a non-power cost law and yield a different abundance shape.

Neither this identity nor the current case studies establishes a universal tendency. This evidential limit differs from concluding that the tendency is absent because many distributions lack its neutral signature. Section 2.5 makes system and medium constraints part of the explanatory programme; the reported tests address specified allocation and response models under their stated conditions. The original linear-allocation assumption and the ecological logarithmic-allocation assumption must be treated separately. The physical examples contradict selected proxy versions, while the ecological summaries contain suggestive locations and substantial uncertainty. The class-exchange result now explains approach to equal allocation under explicit symmetric kinetics, and predicts a constrained equilibrium and recovery. Radial radiation transport derives a physical equality and its medium-dependent departure. Thermal radiation adds classical equality per mode and a quantum departure derived from the allowed states, connected to the published FIRAS measurement with its reconstruction and calibration assumptions. These results do not establish that every natural system follows the same class-exchange mechanism, proves a cosmological time arrow, or quantifies the work needed to change social inequality.

A narrower empirical claim remains possible. It would specify a population of systems, one independently justified resource and measure, a sampling frame, a common or explicitly varying size domain, and an equivalence tolerance with scientific meaning. Raw resource totals would permit testing profile shape as well as slope. A design with multiple sites and repeat visits could separate sources of variation using a model appropriate to its sampling structure. A new dataset could test a frozen conditional prediction, including its uncertainty; numerical proximity to a dispersion fitted on the original sample is not sufficient. Section 5 reports such tests. Their positive results concern budget closure, not neutrality. The plant application additionally shows why normalized snapshots cannot be interchanged with normalized expected stocks without an observation model.

For natural size-class applications, the principal limitation is the absence of independently specified conditions identifying the proposed neutral regime and of data/design sufficient to test its generalization. The physical examples have explicit equality conditions, but those conditions do not automatically transfer to ecological or social concentrations. An empirically supported allocation law need not explain its symmetry through a deeper mechanism. Its predictions must

nevertheless specify the resource, class measure, and statistical target. An experiment that weakens an independently measured source of bias could test approach to the neutral prediction, if that convergence is part of the declared hypothesis. To explain a departure, identify the restriction and predict its effect through a combined model; naming a previously unspecified restriction does not supply that prediction. The tests of Section 5 sharpen this limitation. Budget closure with a geometric cost is confirmed in separately grown lineages that share one medium. A cost-ratio rule for redistribution among coexisting classes failed in a grazed community.

Prior ecological work already supplies models and environmental explanations (Cuesta, Delius and Law, 2018; Arranz et al., 2022). Marshall et al. (2022) independently measured metabolic scaling and predicted maximum density and biovolume in evolved bacteria. Cost-to-capacity prediction therefore has direct experimental precedents. Dillon et al. (2019) already established curvature and superior Weibull fits in the plant data; the present application adds transferred stock/count comparisons and observation limits. Further work should connect resource symmetry to explicit mechanisms, derive constrained outcomes, and test predictions beyond these existing accounts.

7. Conclusion

The neutral allocation form of orthopolity is mathematically precise and testable once its resource, measure, domain, and statistical target are fixed. Its inverse-cost consequence is an accounting identity shared with established size-spectrum theory. Resource symmetry provides a coherent postulate for equal expected allocations; a deeper mechanism is not required for it to become an empirically established law. The broader principle concerns a tendency whose observable expression depends on system and medium constraints. Departures from neutrality therefore address particular formulations and conditions; they do not by themselves imply a general absence of evidence for orthopolity. Establishing the scope of the principle requires both positive demonstrations and predictive explanations of constrained outcomes. The stronger universal, cosmological, and ensemble-confirmation claims remain unestablished by the arguments and evidence presented here. Frozen forecasts support the budget-closure form of the identity in an engineered allocation system and in algal lineages sharing one medium. They reject a two-budget cost-ratio rule for short-term redistribution in grazed food webs. Direct coexisting plant stocks give mixed template transfer and an unresolved expected-neutrality verdict. The class-exchange construction adds a mechanism selecting neutrality and a non-neutral state sustained by opposing contributions, with a proved recovery bound. Electromagnetic transport adds a physical equality with an explicitly resolved absorption constraint and propagation-limited recovery. Thermal radiation adds equal classical mean energy per mode, quantum suppression, and its connection to the published FIRAS spectrum under a declared observation model. Together these results connect the proposed tendency to governing dynamics, constraints, and observable outcomes, while retaining the scope of each derivation and empirical claim.

Data and code availability

Code, frozen inputs, configurations, and generated results are in this repository. The make all target verifies input checksums, runs tests, and regenerates the empirical analyses; see the README for setup. Pilot estimates originate in results.json; distribution diagnostics in gof.json; aquatic median, compatibility fractions, and ocean intervals in independent.json. Remaining JSON files contain explicitly exploratory diagnostics. The studies of Section 5 are registered in an append-only run registry with their inputs, frozen protocols, archived algorithms and outputs. The class-exchange demonstration and retrospective FIRAS calculation are also registered; their

reports give offline reproduction commands and interpretation limits. Plant sources are pinned to author commit defccc3dcbbbf3ba57ff1572377de88fba83ff7f, with acquisition checksums and complete inclusion ledgers. The registry-verify target audits the registry offline, and each study report gives its replay command; source errata found during those audits are recorded there. Source-specific conditions are recorded in data/NOTICE.md; there is no blanket licence for all inputs. The supplied 2017 manuscript was inspected privately and is not redistributed.

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